

Optimization Methods for Machine Learning and Engineering

Lecture 7 – Optimization in Vector Spaces

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Agenda

1. Vector Spaces
2. Norms and Banach Spaces
3. Inner Products, Hilbert Spaces and the Projection Theorem
4. Applications

Vector Spaces

Vector Spaces

A **set of elements** X with the operations

Addition: $\forall x, y \in X, x + y \in X$

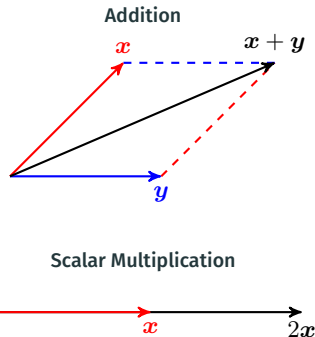
Scalar Multiplication: $\forall x \in X, \alpha \in \mathbb{R}, \alpha x \in X$

is called a **vector space** if in addition the following **axioms** are fulfilled for any elements $x, y, z \in X$ and scalars $\alpha, \beta \in \mathbb{R}$:

- | | |
|---|--------------------|
| 1. $x + y = y + x$ | (Commutative Law) |
| 2. $(x + y) + z = x + (y + z)$ | (Associative Law) |
| 3. $(\alpha\beta)x = \alpha(\beta x)$ | (Associative Law) |
| 4. $\alpha(x + y) = \alpha x + \alpha y$ | (Distributive Law) |
| 5. $(\alpha + \beta)x = \alpha x + \beta x$ | (Distributive Law) |
| 6. $\exists 0 \in X$ such that $x + 0 = x, \quad \forall x \in X$ | (Null Vector) |
| 7. $0x = 0, \quad 1x = x$ | |

The elements of X could be from $\mathbb{R}^n$, keeping with our previous notion of a “vector”. But many **other types of mathematical objects** also form vector spaces. And **not all** $X \subset \mathbb{R}^n$ **obey the axioms**.

The operations on vectors from $\mathbb{R}^n$

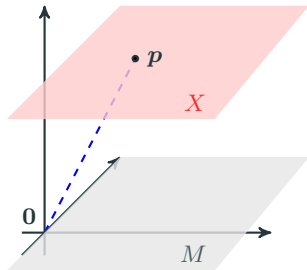


Quiz: Is X a Vector Space?

$$X = \mathbb{R}^n \text{ with } n \in \mathbb{N}$$

Yes

Quiz: Is X a Vector Space?



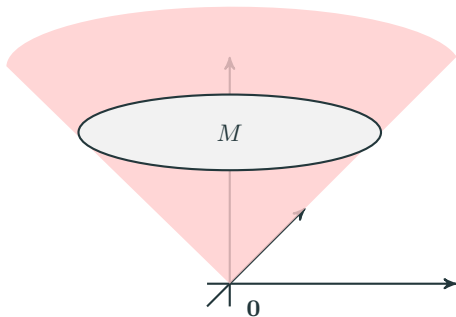
$$M = \{\mathbf{m} \in \mathbb{R}^3 \mid m_3 = 0\}$$

$$\mathbf{p} \in \mathbb{R}^3$$

$$X = \{\mathbf{m} \in M + \mathbf{p}\}$$

No

Quiz: Is X a Vector Space?

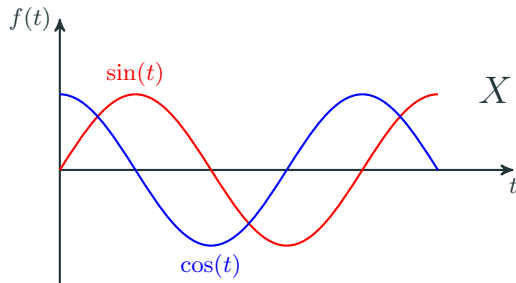


$$M \subseteq \mathbb{R}^n \text{ with } n \in \mathbb{N}$$

$$X = \{ \mathbf{x} \mid \exists \mathbf{m} \in M, \alpha \geq 0 \\ \mathbf{x} = \alpha \mathbf{m} \}$$

No

Quiz: Is X a Vector Space?



$$X = \{f \mid \exists \alpha, \beta \in \mathbb{R}, \\ f = t \mapsto \alpha \sin(t) + \beta \cos(t)\}$$

Yes

For the addition of functions use $(f + g)(t) = f(t) + g(t)$.

The null vector $\mathbf{0}$ of function-space is $f_0(t) = 0$.

Linear Combinations, Linear Dependence, Basis and Dimensions

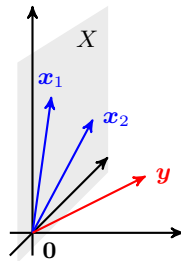
A vector x from a vector space X is a **linear combination** of the vectors $\{y_1, y_2, \dots\} \subseteq X$ if there exist scalars $\{\alpha_1, \alpha_2, \dots\}$ so that $x = \sum_i \alpha_i x_i$. Note that we could have infinitely many y_i for the linear combination.

The vectors $\{x_1, \dots, x_n\}$ from a vector space X are **linearly independent** if $\sum_i \alpha_i x_i = \mathbf{0}$ implies $\alpha_i = 0$ for all i .

A set of linearly independent vectors $\{x_1, x_2, \dots\}$ is called a **basis** of X if its linear combinations span X .

The **dimension** of a vector space X is defined by the number of elements in its basis.

We first encountered these concepts in the context of Linear Algebra in $\mathbb{R}^n$. But they are **more general and can be applied to any vector space**. This is a common theme for this lecture.



- Vectors x_1 and x_2 are a basis for the vector space X
- y is linearly independent from x_1 and x_2 and is therefore not an element of X

Norms and Banach Spaces

Normed Vector Spaces

A **normed vector space** additionally has a real-valued function that maps each element $x \in X$ into a **real number** $\|x\|$ **called the norm of x** where the following axioms hold:

1. $\|x\| \geq 0$, $\|x\| = 0$ iff $x = 0$
2. $\|x + y\| \leq \|x\| + \|y\| \quad \forall x, y \in X$ (Triangle Inequality)
3. $\|\alpha x\| = |\alpha| \cdot \|x\| \quad \forall \alpha \in \mathbb{R}$

Every norm implies a **metric**, i.e. a distance function d between vectors $x, y \in X$:

$$d(x, y) := \|x - y\|$$

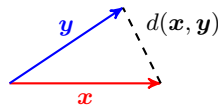
There, from the norm axioms, we have

1. $d(x, y) = 0 \Leftrightarrow x = y$ (Identity of Indiscernibles)
2. $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle Inequality)
3. $d(x, y) = d(y, x)$ (Symmetry)

Norm of a vector as its length



Distance between vectors



The p -Norms

For elements $\mathbf{x} \in \mathbb{R}^n$, the previously encountered **Euclidean Norm** $\|\cdot\|_2$ is only a special case from the family of p -Norms

$$\|\mathbf{x}\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

for $p \geq 1$. Other common values for p are:

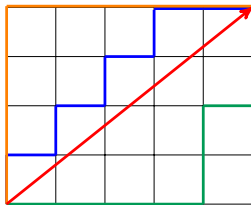
$p = 1$ The **Manhattan Norm** is simply the sum of the absolute values.

$p = \infty$ The **Maximum Norm** arises in the limit when p is increased. It can be defined alternatively as

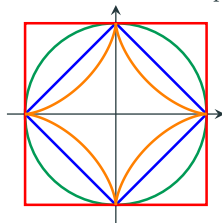
$$\|\mathbf{x}\|_\infty = \max_i |x_i|.$$

In the example on the right-hand side, there is a unique shortest path in the Euclidean distance (implied by the Euclidean Norm) across the grid (red). In Manhattan distance there are several paths with the same length.

Distances in the Manhattan Norm ($p = 1$)



Unit circle for different p



$p = \infty$, $p = 2$, $p = 1$, $p = 1/2$

Convergence and Banach Spaces

In the context of open/closed sets, we previously saw a **convergent sequence**. Now we can make this notion of convergence more precise.

Let $\{x_i\} \subseteq X$ an infinite series from the normed vector space X . The series converges if there exists some element $y \in X$ for which $\|y - x_i\|$ converges to zero. More precisely, for every $\varepsilon > 0$ there exists an index m such that $\|y - x_i\| < \varepsilon$ for all $i \geq m$. We then write $x_i \rightarrow y$.

A sequence $\{x_i\}$ is said to be a **Cauchy sequence** if $\|x_i - x_j\| \rightarrow 0$ as $i, j \rightarrow \infty$; i.e., given $\varepsilon > 0$, there is an index m such that $\|x_i - x_j\| < \varepsilon$ for all $i, j \geq m$.

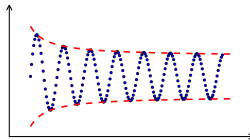
In a normed space every convergent sequence is a Cauchy sequence.

A normed vector space X is **complete** if every Cauchy sequence from X has a limit in X . A complete normed vector space is called a **Banach space**.



Stefan Banach (1892 – 1945)

A non-Cauchy sequence



Completeness and the existence of Fixed Points

In a normed vector space, **any finite-dimensional subspace is complete**. So all normed vector spaces embedded in $\mathbb{R}^n$ are Banach spaces.

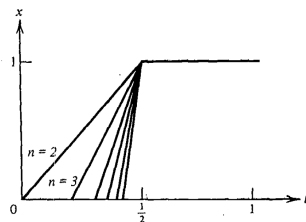
Completeness is a prerequisite for many of the optimization algorithms we saw prior. For example, to show convergence of Gradient Descent and the Newton Method in general normed vector spaces.

Let S be a subset of a normed vector space X and let f be a transformation $f : S \rightarrow S$. Then f is a **contraction** if there exists an α with $0 \leq \alpha < 1$ such that $\|f(x) - f(y)\| \leq \alpha\|x - y\|$ for all $x, y \in S$.

Banach Fixed Point Theorem

If f is a contraction on a closed subset S of a Banach space, there is a fixed point $x^* \in S$ satisfying $x^* = f(x^*)$. Furthermore, x^* can be obtained by starting with an arbitrary $x_0 \in S$ and following a sequence $x_{i+1} = f(x_i)$.

A Non-Complete Normed Space [Luenberger69]



Consider the normed vector space of continuous functions $L^2[0, 1]$. Let a sequence of functions f_i from this space:

$$f_i(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq \frac{1}{2} - \frac{1}{i} \\ it - \frac{i}{2} + 1 & \text{for } \frac{1}{2} - \frac{1}{i} \leq t \leq \frac{1}{2} \\ 1 & \text{for } t \geq \frac{1}{2} \end{cases}$$

Each function f_i is continuous for finite i . However the sequence converges in the limit to the step function which is not continuous and not in $L^2[0, 1]$.

Inner Products, Hilbert Spaces and the Projection Theorem

The Inner Product

Let X a vector space. The **inner product** $\langle x | y \rangle$ is a function defined on $X \times X$ that maps each pair of vectors $x, y \in X$ to a scalar while satisfying the following axioms:

1. $\langle x + y | z \rangle = \langle x | z \rangle + \langle y | z \rangle$ (Linearity in the first argument)
2. $\langle \lambda x | y \rangle = \lambda \langle x | y \rangle$ (Linearity in the first argument)
3. $\langle x | y \rangle = \overline{\langle y | x \rangle}$ (Conjugate Symmetry)
4. $\langle x | x \rangle \geq 0$ and $\langle x | x \rangle = 0$ iff $x = 0$ (Positive Definiteness)

The $\overline{}$ denotes complex conjugation (complex-valued vector spaces are not considered in the course).

A vector space with an inner product defined is a **pre-Hilbert space**.

Every **inner product implies a norm** $\|x\| = \sqrt{\langle x | x \rangle}$.

Euclidean Inner Product

A vector space $X \subseteq \mathbb{R}^n$ with elements x, y and the inner product

$$\langle x | y \rangle = \sum_{i=1}^n x_i y_i .$$

Function Spaces

The vector space $L^2[a, b]$ of continuous functions f, g with $\int_a^b f(t)^2 dt < \infty$ and the inner product

$$\langle f | g \rangle = \int_a^b f(t) g(t) dt .$$

Orthogonality and the Projection Theorem

Two elements x, y from a pre-Hilbert space are said to be **orthogonal** if $\langle x | y \rangle = 0$, denoted as $x \perp y$.

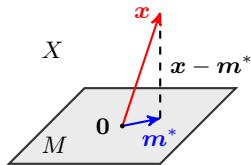
If x, y are orthogonal $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

Proof: $\|x + y\|^2 = \langle x + y | x + y \rangle = \langle x | x + y \rangle + \langle y | x + y \rangle = \langle x + y | x \rangle + \langle x + y | y \rangle = \langle x | x \rangle + \langle y | x \rangle + \langle x | y \rangle + \langle y | y \rangle = \|x\|^2 + \|y\|^2$

Consider the following optimization problem: Let a pre-Hilbert space X and a subspace $M \subset X$. Given an element $x \in X$, what is the element $m \in M$ that minimizes $\|x - m\|$?

Projection Theorem for pre-Hilbert Spaces see [Luenberger69]

If there is an element $m^* \in M$ such that $\|x - m^*\| \leq \|x - m\|$ for all $m \in M$, then m^* is unique. The element m^* is a unique minimizer in M iff the residual $x - m^*$ is orthogonal to M .

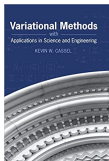
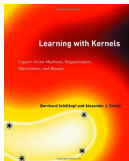


A complete pre-Hilbert space is called a **Hilbert space**.

Concerning the Projection Theorem, we know that a **unique minimizer must exist for Hilbert spaces**.

Results from Linear Algebra are generalized to infinite-dimensional Vector Spaces. **Linear Operators** translate between different Hilbert Spaces. Matrix multiplication is a special case for linear operators in the finite-dimensional case.

Hilbert Spaces are used in many different fields



John Von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, 1932

Bernhard Schölkopf and Alexander J Smola. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. MIT press, 2002

Kevin W Cassel. *Variational methods with applications in science and engineering*. Cambridge University Press, 2013



David Hilbert (1862 – 1943)

The last person who knew all of mathematics (Folklore)

Gram-Schmidt-Orthogonalization

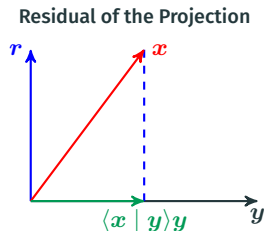
In an **orthogonal set** S all elements are mutually orthogonal $\forall x, y \in S$, $x \neq y \Rightarrow x \perp y$.

If S is **orthonormal** (in addition to orthogonal), then $\forall x \in S$, $\|x\| = 1$.

Given $x, y \in X$ and $\|y\| = 1$, then $\langle x | y \rangle y$ is the projection of x on y .

The **residual of the projection** $r = x - \langle x | y \rangle y$ is orthogonal to y .

Proof: $\langle x - \langle x | y \rangle y | y \rangle = \langle x | y \rangle - \langle x | y \rangle \langle y | y \rangle = 0$



Let $\{b_1, b_2, \dots, b_n\}$ a finite basis for the subspace M of a Hilbert space $H \supseteq M$. We can construct an orthonormal basis $\{e_1, e_2, \dots, e_n\}$ for M using **Gram-Schmidt-Orthogonalization**:

$$e_1 = \frac{b_1}{\|b_1\|}, \quad e_n = \frac{b_n - \sum_{i=1}^{n-1} \langle b_n | e_i \rangle e_i}{\|b_n - \sum_{i=1}^{n-1} \langle b_n | e_i \rangle e_i\|}$$

By the Projection Theorem we find $m^* \in M$ with minimum distance to some $x \in H$ as

$$m^* = \arg \min_{\alpha_1, \alpha_2, \dots, \alpha_n} \|x - \sum_{i=1}^n \alpha_i b_i\| = x - \sum_{i=1}^n \langle x | e_i \rangle e_i$$

The Normal Equations

Again, we look at the minimum norm projection $\mathbf{m}^* = \arg \min_{\alpha_1, \alpha_2, \dots, \alpha_n} \|\mathbf{x} - \sum_{i=1}^n \alpha_i \mathbf{b}_i\|$ where the $\mathbf{b}_i$ span a subspace of a Hilbert space H . But instead of just $\mathbf{m}^*$ **we are also interested in the α_i** . Gram-Schmidt-Orthogonalization does not immediately give us those.

From the Projection Theorem we know that the residual $\mathbf{x} - \sum_{i=1}^n \alpha_i \mathbf{b}_i$ is orthogonal to all $\mathbf{b}_i$.

$$\begin{aligned}\langle \mathbf{x} - \sum_{i=1}^n \alpha_i \mathbf{b}_i \mid \mathbf{b}_i \rangle &= 0, & \forall i = 1, \dots, n \\ \langle \sum_{i=1}^n \alpha_i \mathbf{b}_i \mid \mathbf{b}_i \rangle &= \langle \mathbf{x} \mid \mathbf{b}_i \rangle, & \forall i = 1, \dots, n\end{aligned}$$

We can further unpack the left-hand side to get a system $\boxed{\mathbf{G}\boldsymbol{\alpha} = \mathbf{c}}$ of **n linear equations with n unknowns**. These are known as the **Normal Equations**. Note that only $\mathbf{c}$ depends on the vector $\mathbf{x}$ that we want to project.

$$\begin{array}{ccccccc}\langle \mathbf{b}_1 \mid \mathbf{b}_1 \rangle \alpha_1 & + & \langle \mathbf{b}_2 \mid \mathbf{b}_1 \rangle \alpha_2 & + & \dots & + & \langle \mathbf{b}_n \mid \mathbf{b}_1 \rangle \alpha_n & = & \langle \mathbf{x} \mid \mathbf{b}_1 \rangle \\ \langle \mathbf{b}_1 \mid \mathbf{b}_2 \rangle \alpha_1 & + & \langle \mathbf{b}_2 \mid \mathbf{b}_2 \rangle \alpha_2 & + & \dots & + & \langle \mathbf{b}_n \mid \mathbf{b}_2 \rangle \alpha_n & = & \langle \mathbf{x} \mid \mathbf{b}_2 \rangle \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ \langle \mathbf{b}_1 \mid \mathbf{b}_n \rangle \alpha_1 & + & \langle \mathbf{b}_2 \mid \mathbf{b}_n \rangle \alpha_2 & + & \dots & + & \langle \mathbf{b}_n \mid \mathbf{b}_n \rangle \alpha_n & = & \langle \mathbf{x} \mid \mathbf{b}_n \rangle\end{array}$$

The Gram Matrix

Let $\{b_1, b_2, \dots\}$ a linearly independent basis from a Hilbert space. Its **Gram matrix** can be precomputed as

$$\mathbf{G}(b_1, b_2, \dots, b_n) = \begin{pmatrix} \langle b_1 | b_1 \rangle & \langle b_2 | b_1 \rangle & \dots & \langle b_n | b_1 \rangle \\ \langle b_1 | b_2 \rangle & \langle b_2 | b_2 \rangle & \dots & \langle b_n | b_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle b_1 | b_n \rangle & \langle b_2 | b_n \rangle & \dots & \langle b_n | b_n \rangle \end{pmatrix}.$$

Theorem: The determinant of the Gram matrix is non-null $|\mathbf{G}(b_1, b_2, \dots, b_n)| \neq 0$ iff the b_i are linearly independent.

In that case, the matrix is invertible and we can solve $\mathbf{G}\alpha = c$ for α with standard methods.

Hence, for every finite basis embedded in a Hilbert space, we can compute the minimum distance projection and express it by coefficients α_i for the basis elements.

Minimum Distance in Julia

```
# Norm and distance function
norm(L, x) = sqrt(inner(L, x, x))
dist(L, x, y) = norm(L, x-y)

# Example for the Euclidean p2-Norm
inner(::Val{:P2}, x, y) = x' * y
dist(Val{:P2}, [0,0], [1,1]) # 1.4142
```

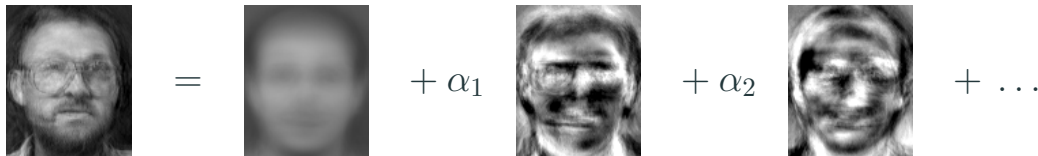
```
function gram_schmidt(L, b)
    e = copy(b)
    for i = 1:length(b)
        for j=1:i-1
            e[i] -= e[j] * inner(L, b[i], e[j])
        end
        nn = norm(L, e[i])
        if nn > 0.0001 # Normalize if non-zero
            e[i] = e[i] / nn
        end
    end
    return e
end
```

```
# Projection on the subspace defined by a (not
# necessarily orthogonal) basis
function proj(L, x, basis)
    ob = gram_schmidt(L, basis) # orthonormal basis
    return sum([ob[i] * inner(L, x, ob[i])
                for i=1:length(ob)])
end

# Returns the projection and its coefficients
# for the basis elements
function proj_normal(L, x, basis)
    nb = length(basis)
    G = zeros(nb,nb) # Gram matrix, always symmetric
    for i=1:nb, j=1:i
        G[i,j] = inner(L, basis[i], basis[j])
        G[j,i] = G[i,j]
    end
    c = [inner(L, x, basis[i]) for i=1:nb]
    alpha = G \ c # G * alpha = c
    return sum(basis .* alpha), alpha
end
```

Applications

Catching Bad Guys with Eigenfaces


$$\text{Target Face} = \text{Average Face} + \alpha_1 \text{Eigenface}_1 + \alpha_2 \text{Eigenface}_2 + \dots$$

- From a database of face images, compute the “average face” and n **Eigenfaces**.
 - The Eigenfaces are extracted using the Eigen-decomposition technique already encountered for Fibonacci-in-constant-time (not further discussed here).
 - The **Eigenfaces are a basis** for a (finite) n -dimensional **vector space**.
- For every face image, we can find a **minimum-distance projection on the face-space**. This gives us n -dimensional coefficients α that we can use as features.
- Recognize a person by **nearest-neighbor lookup for the Eigenface coefficients** of known faces.

Lawrence Sirovich and Michael Kirby. “Low-dimensional procedure for the characterization of human faces”. In: *Journal of the Optical Society of America A* 4.3 (1987), pp. 519–524

Approximating $\sin$ with a Polynomial

The vector space $L^2[0, 1]$ contains **continuous functions** $\cdot : [0, 1] \rightarrow \mathbb{R}$

- with the inner product $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$ and the corresponding
- norm $\|f\| = \sqrt{\int_0^1 f(t)^2 dt}$ (restrict to f where $\|f\| < \infty$).

Let the vector space $P_n \subset L^2[0, 1]$ with the **polynomials of n th degree**.

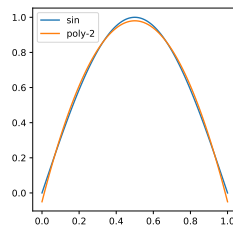
- A polynomial of n th degree can be represented by an $(n + 1)$ -vector of its coefficients (including the intercept).
- Any set of polynomial functions spans a subspace of $L^2[0, 1]$. We can compute an orthonormal basis for it.

With this, we can perform a **minimum-distance projection** from the **continuous functions on the polynomials** of n th degree.

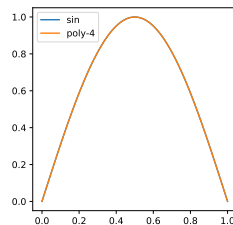
Which polynomial of n th degree most closely represents $g(t) = \sin(\pi t)$?

Solve as a **minimum norm problem** $f_n = \arg \min_{h \in P_n} \|h - g\|$.

$$f_2(t) \approx -0.050 + 4.121t - 4.121t^2$$



$$f_4(t) \approx 0.001 + 3.087t + 0.536t^2 - 7.247t^3 + 3.623t^4$$



Approximating $\sin$ with a Polynomial in Julia

```
import Base: +,-,*,/

# Polynomials representation and evaluation
struct Poly
    c::Vector{Float64} # Coefficients (intercept 1st)
end

(f::Poly)(x) = sum([x^(i-1)*f.c[i] for i=1:length(f.c)])

# Addition and subtraction
+(f::Poly, g::Poly) = Poly(f.c .+ g.c)
-(f::Poly, g::Poly) = Poly(f.c .- g.c)

# Multiplication and division with a real scalar
*(f::Poly, y::T) where T <: Real = Poly(f.c * y)
/(f::Poly, y::T) where T <: Real = Poly(f.c / y)

# Examples
pp = Poly([1,2,0])
pp(2.0) # 1 + 2 * 2.0 + 0 * 2.0^2 = 5.0

pp2 = pp*2 + Poly([1,1,1])
pp2(1.5) # 3 + 5 * 1.5 + 1 * 1.5^2 = 12.75
```

```
# Inner product for functions from L2[0,1]
function inner(::Val{L2}, f, g)
    dt = 0.001 # Approximate the integral
    return sum([f(t)*g(t)*dt for t=0.0:dt:1.0])
end

# Project sin on the second degree polynomials
g(t) = sin(t*pi)
p_basis = [Poly([1,0,0]),
            Poly([0,1,0]),
            Poly([0,0,1])]
g_proj, g_coeff = proj_normal(Val{L2}, g, p_basis)
# g_coeff = [-0.05016328783041,
#            4.12100032032210,
#            -4.12100032032211]

# How is the sine function actually computed
# by the OS / standard math library (libm)?
# - http://www.netlib.org/fdlibm/k\_sin.c
# - http://www.netlib.org/fdlibm/s\_sin.c
# Or via CORDIC algorithms in hardware
```

Quadratic Optimization with Equality Constraints

$$\begin{aligned} \mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \mathbf{x}^\top \mathbf{Q} \mathbf{x} \\ \text{subject to} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned}$$

All solutions fulfilling the equality constraint lie in a **linear variety** V . Linear varieties contain elements from some vector space with an additional offset away from the null vector.

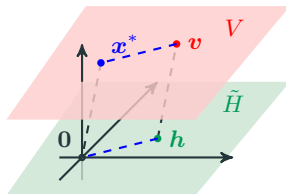
Note that $\langle \mathbf{x} \mid \mathbf{x} \rangle_{\mathbf{Q}} = \mathbf{x}^\top \mathbf{Q} \mathbf{x}$ is a valid inner product. Which element of V is closest to $\mathbf{0}$ wrt. the implied distance metric?

1. Find some $\mathbf{v}$ that fulfills the constraint $\mathbf{A} \mathbf{v} = \mathbf{b}$.
2. Let the Hilbert Space $\tilde{H}$ the **nullspace of $\mathbf{A}$** with the inner product $\langle \cdot \mid \cdot \rangle_{\mathbf{Q}}$. $\tilde{H}$ is parallel to V . Project $\mathbf{v}$ onto $\tilde{H}$:

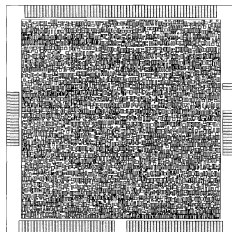
$$\mathbf{h} = \arg \min_{\mathbf{g} \in \tilde{H}} \langle \mathbf{v} \mid \mathbf{g} \rangle_{\mathbf{Q}}$$

3. The solution is $\mathbf{x}^* = \mathbf{v} - \mathbf{h}$.

Projection with Equality Constraints



Application Example: Sea-of-Gates VLSI Optimization [Kleinhans1991]



Conjugate Gradient (CG)

Similar to Gradient Descent, but with an additional processing step for the gradient [Hestenes1952; Hestenes1980].

First step direction: $\mathbf{d}^{(1)} = -\nabla f(\mathbf{x}^{(0)})$

Later step directions:

1. Start with $\tilde{\mathbf{d}}^{(k)} = -\nabla f(\mathbf{x}^{(k-1)})$.
2. Compute $\mathbf{d}^{(k)}$ by orthogonalization of $\tilde{\mathbf{d}}^{(k)}$ wrt. the previous step directions $\{\mathbf{d}^{(1)}, \dots, \mathbf{d}^{(k-1)}\}$.
3. Additional linesearch (specialized linesearch methods for CG exist).

For an unconstrained quadratic optimization problem in n dimensions, Conjugate Gradient converges within n steps.

The Newton method would solve it in one step. But with the added cost of computing the Hessian and solving a linear equation for it.

Note that Hestenes et al. [developed CG on a Zuse Z4 computer](#).

Conjugate Gradient for a
Quadratic Objective

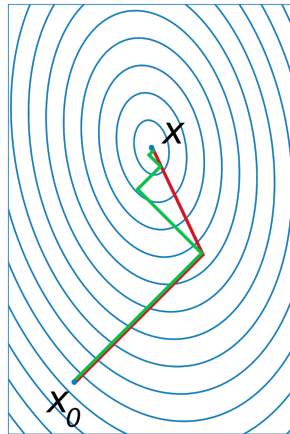


Image Source: Wikipedia

Summary of what you learned today

- **Vector spaces** and their axioms
- **Banach spaces** and norms beyond Euclidean distances
- **Hilbert spaces** and inner products with a notion of **orthogonality**
- Computing an orthonormal basis with the **Gram-Schmidt Algorithm**
- **Minimum-Norm Projection** on the subspace of a Hilbert space via the **Normal Equations**
- **Applications** for Minimum-Norm Projection
 - Catching Bad Guys with Eigenfaces
 - Approximating the sine function with a polynomial
 - Quadratic Optimization with Equality Constraints
 - Conjugate Gradient

That's it for today.

See you next week for Lecture 8 on
Duality

- [Cassel2013] Kevin W Cassel. *Variational methods with applications in science and engineering*. Cambridge University Press, 2013.
- [Hestenes1980] Magnus Rudolph Hestenes. *Conjugate direction methods in optimization*. Springer, 1980.
- [Hestenes1952] Magnus R Hestenes, Eduard Stiefel, et al. “Methods of conjugate gradients for solving linear systems”. In: *Journal of research of the National Bureau of Standards* 49.6 (1952), pp. 409–436.
- [Kleinhans1991] Jürgen M Kleinhans et al. “GORDIAN: VLSI placement by quadratic programming and slicing optimization”. In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 10.3 (1991), pp. 356–365.
- [Luenberger69] David G Luenberger. *Optimization by Vector Space Methods*. John Wiley & Sons, 1969.
- [Sirovich1987] Lawrence Sirovich and Michael Kirby. “Low-dimensional procedure for the characterization of human faces”. In: *Journal of the Optical Society of America A* 4.3 (1987), pp. 519–524.
- [Schölkopf2002] Bernhard Schölkopf and Alexander J Smola. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. MIT press, 2002.
- [VonNeumann1932] John Von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, 1932.